Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 100 epochs with a batch size of 128. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the test accuracy during training.

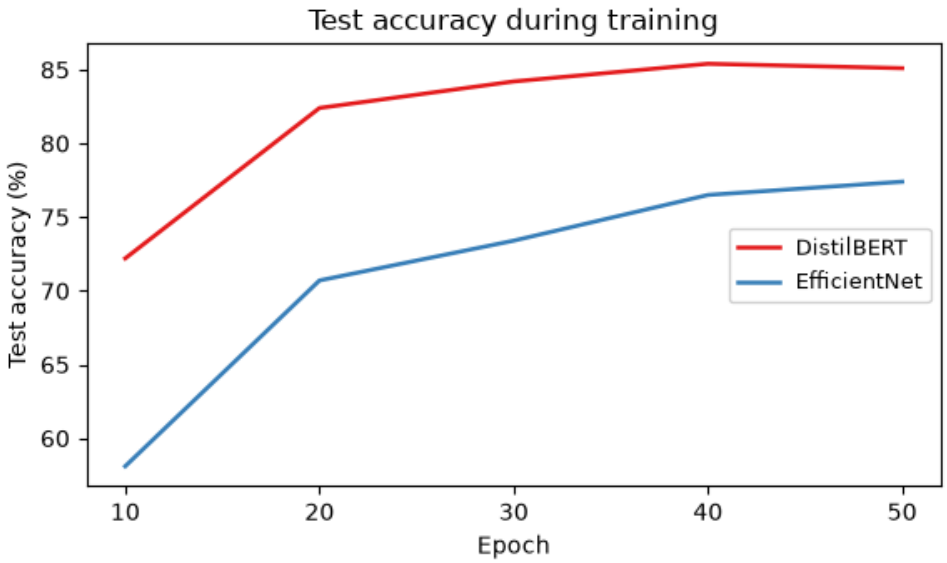

Figure 1: Test accuracy during training for the models in this experiment.

As shown in Figure 1, at epoch 10, EfficientNet achieves 58% test accuracy. As shown in Figure 1, DistilBERT outperforms EfficientNet by 10.8 percentage points at epoch 30. From epoch 30 to epoch 50, the test accuracy of EfficientNet improves by a relative 25.9%. At epoch 30, DistilBERT is 9.3 points better than EfficientNet.

Figure 2 shows the validation accuracy of MLP by dataset.

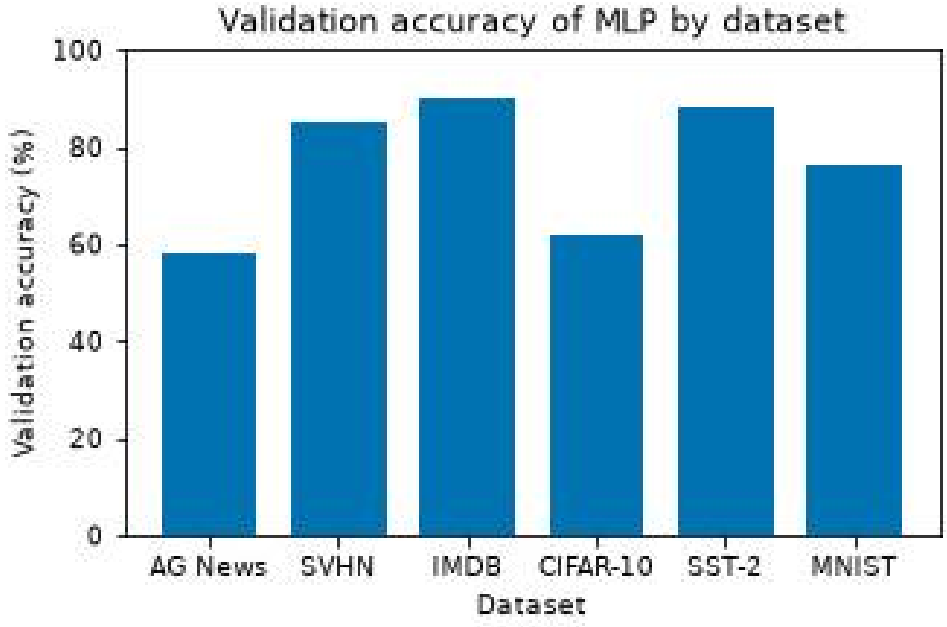

Figure 2: Validation accuracy of MLP by dataset for the models in this experiment.

On MNIST, MLP obtains a validation accuracy of 76%. MLP scores 8.7 points higher on IMDB than on

MNIST (Figure 2). MLP scores 3.1 points higher on SST-2 than on SVHN (Figure 2). On CIFAR-10, MLP obtains a validation accuracy of 89.8% (Figure 2).

3 Discussion

Logistic Regression reaches 69.6% accuracy on QQP. We note that the training was stable for all models.